

Internet of Things

Computer Science Master: IoT Curriculum

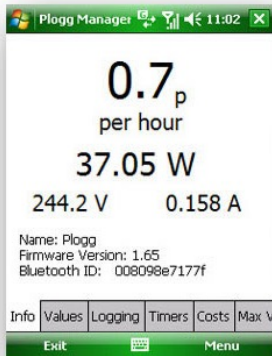
Introduction to the IoT ecosystem

Communication networks are changing...

- Extensions
 - More nodes, more connections
 - Any **TIME**, Any **PLACE** + Any **THING** (**Ubiquity** and **Pervasivity**)
 - M2M communications
 - Billions of interconnected devices,
 - Everybody is connected.
- Expansions
 - Broadband
 - LTE, 5G
- Enhancements
 - Smart networks
 - Data-centric and content-oriented networking
 - Context-aware (autonomous) systems

Sensor devices are becoming widely available...

- Programmable devices
- Off-the-shelf gadgets/tools



Linker Intel Group



Image Sensor Device



But also everyday objects can become sensors ...

My pulse is ...



My blood glucose is ...



My temperature is ...



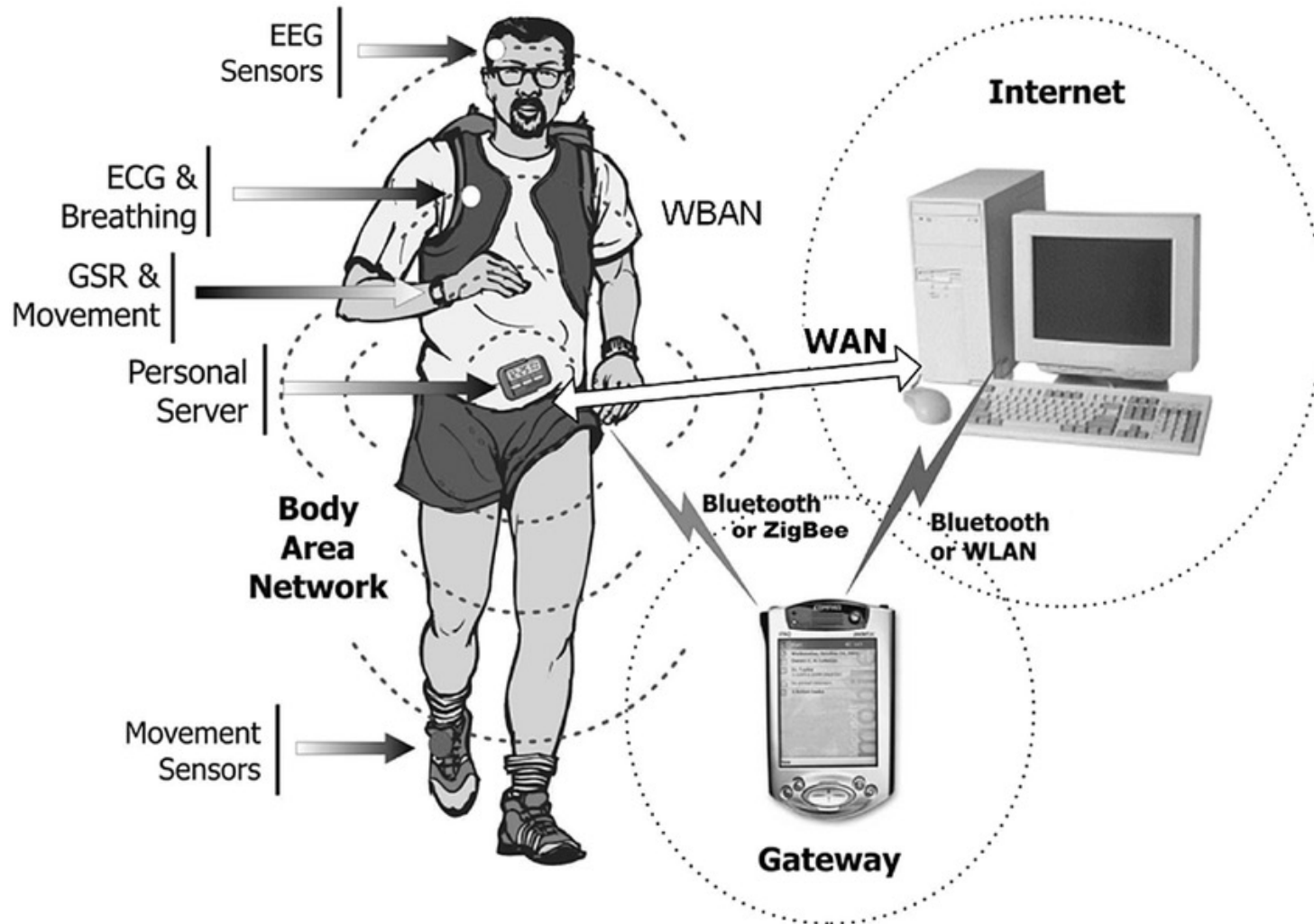
... and monitor things unobtrusively

More “Things” are being connected...

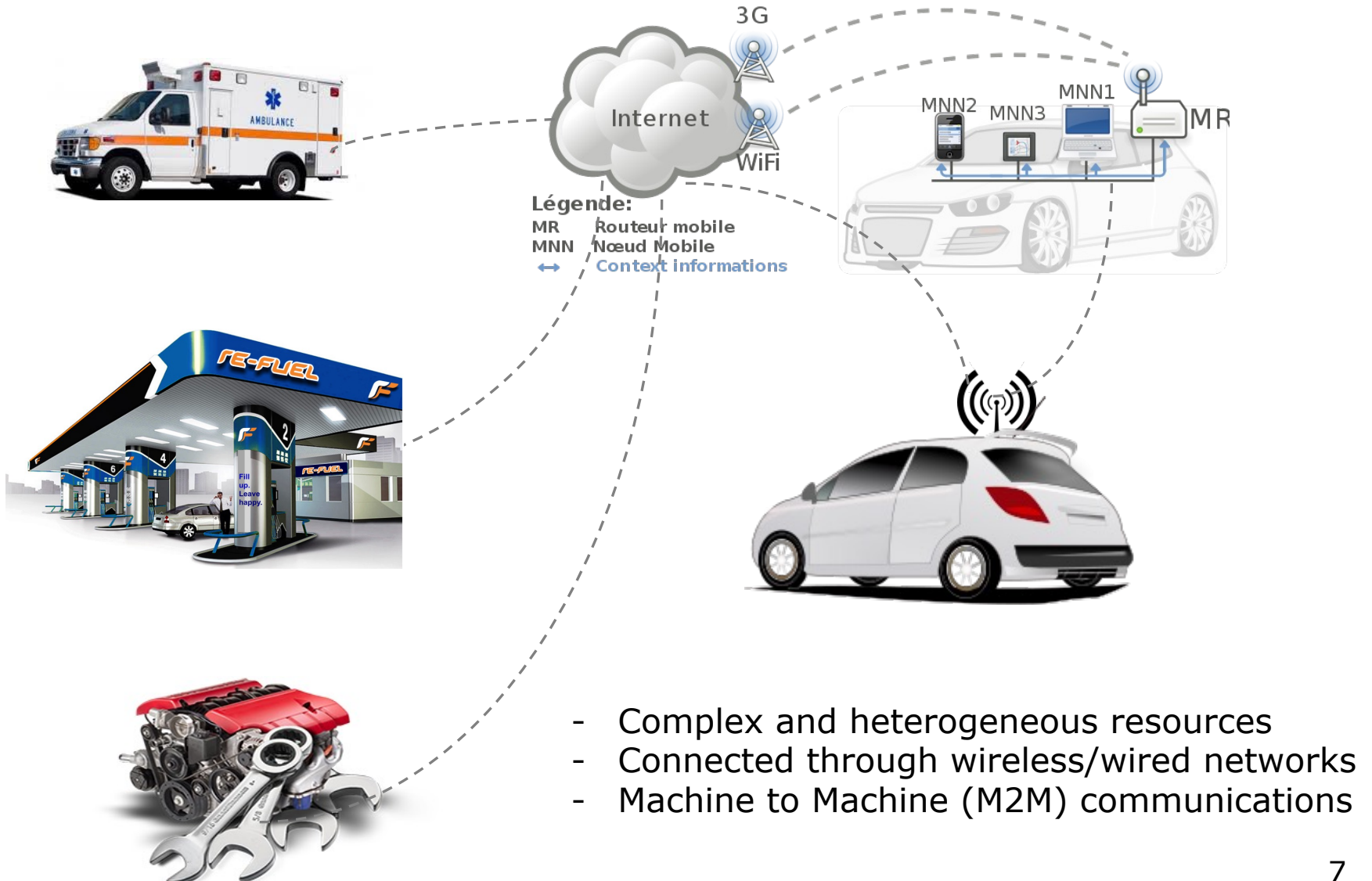
Home/daily-life devices
Business and
Public infrastructure
Health-care
...



People Connecting to Things...



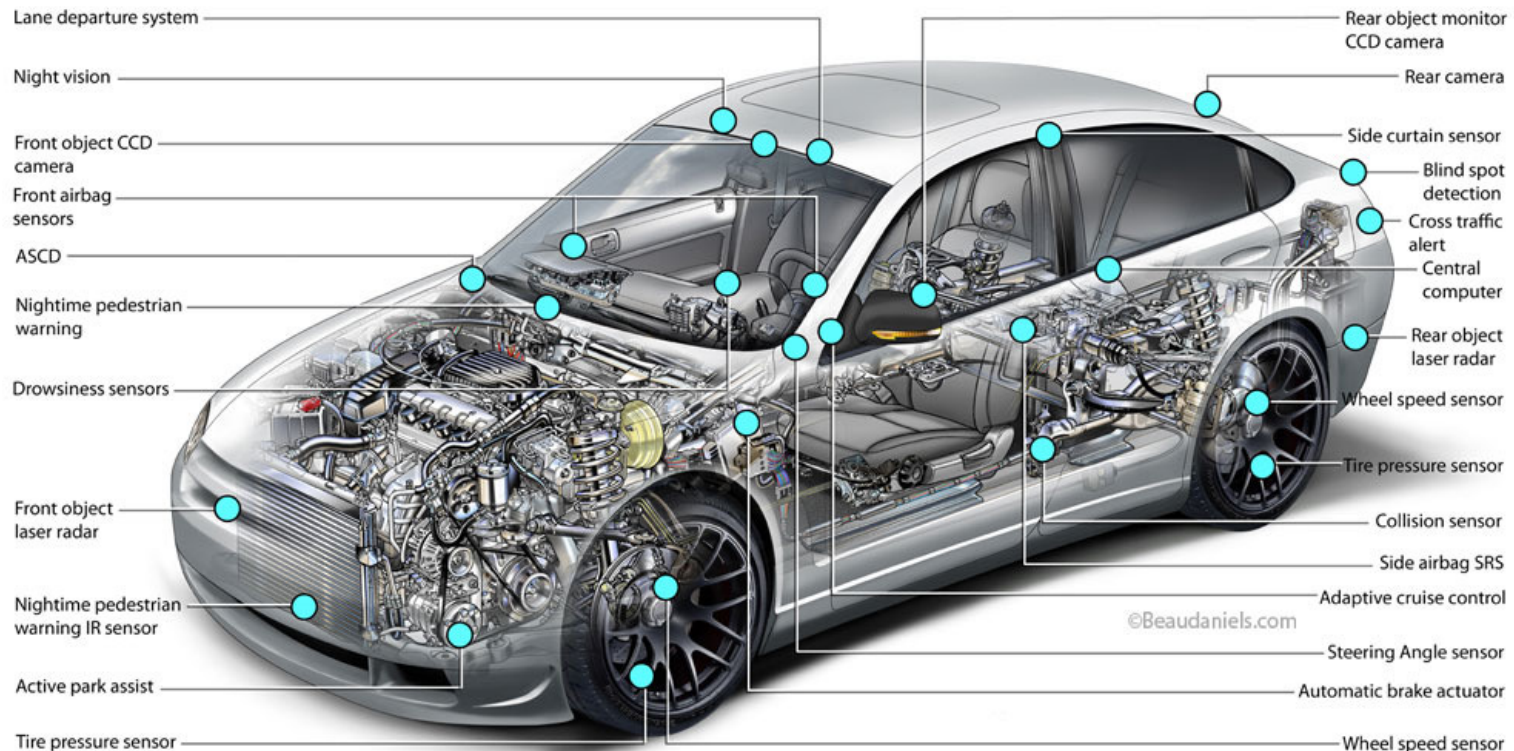
Things Connecting to Things...



An example: modern cars

- economy car: more than 200 sensors (brakes, belts, air bags, doors, etc.)
- luxury car: more than 600 sensors (A/C, ABS, lights, radar, road conditions)
- Vehicle safety and comfort has improved via sensors and actuators

Vehicle Sensors



Another example: Google glasses

- The amazing internet connected glasses that contain a camera, microphone, display and will generally revolutionize the wearable device market



Another example: Luggage tracking

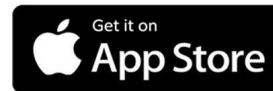
- monitor your luggage on air travel and text you when it arrives at the destination as well as how far it is from the luggage belt.



Another example: WiFi light bulbs

- energy efficient and wifi enabled light bulb that you can control from your smartphone – you can even change colours and mood

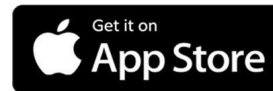
WiFi Smart Bulb



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WiFi Smart Bulb



Another example: electronic fork

- Electronic Bluetooth connected fork that helps you monitor and track your eating habits

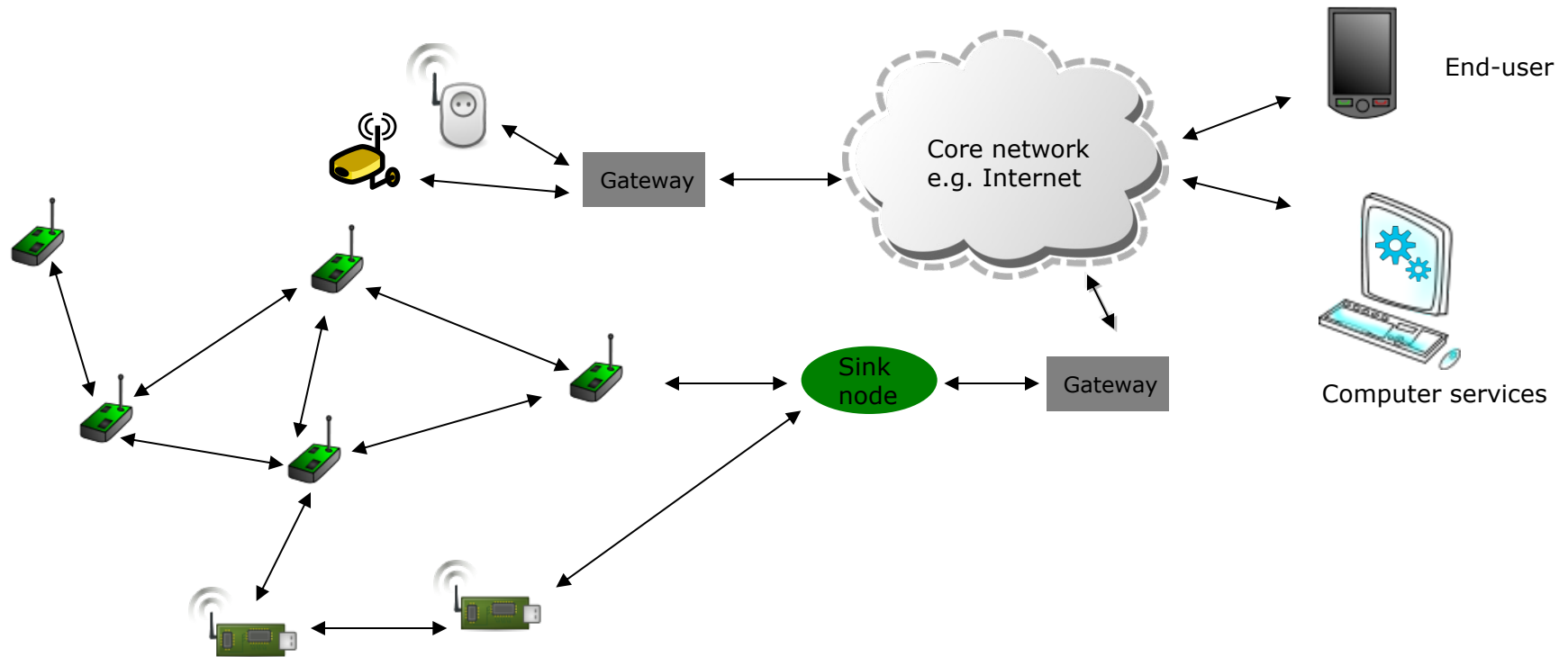


Another example: electronic thermostat

- A thermostat that will get to know you and learn from the way it is used to automatically adjust the temperature in your home



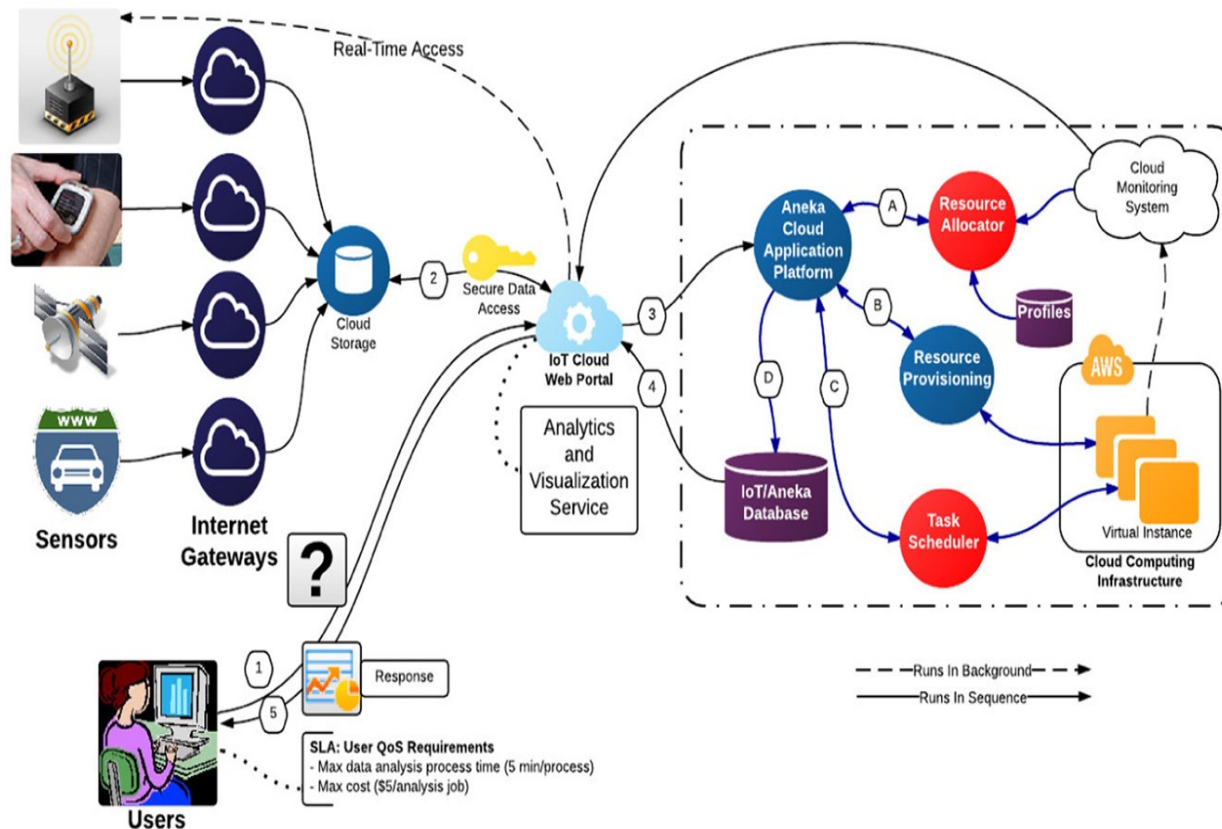
From Wireless Sensor Networks (WSN)...



- The networks typically run Low Power Devices
- Consist of one or more sensors, could be different type of sensors (or actuators)
- Structured into clusters or hierarchies where cluster heads collect and preprocess sensor data
- Cluster heads connect to the external networks through specific gateways

... to the Internet of Things (IoT)

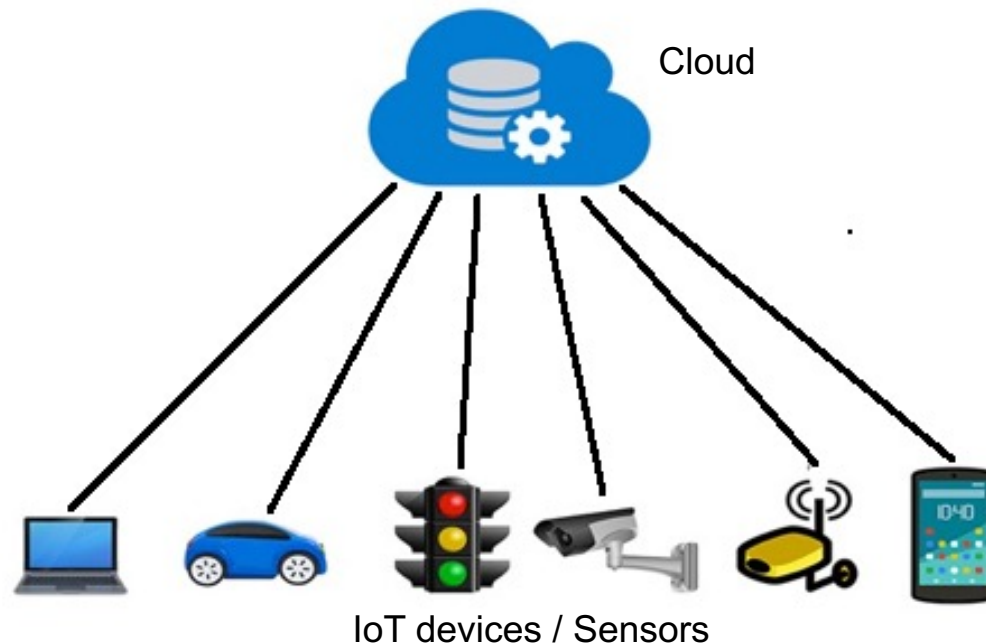
- Extending the current Internet and providing connection, communication, and inter-networking between devices and physical objects, or "Things," is a growing trend that is often referred to as the **Internet of Things (IoT)**.
- Bringing the intelligence into the **cloud** over the Internet



From The traditional Cloud based model...

Cloud solutions provide unlimited computing power and storage space to hardware-constrained IoT devices.

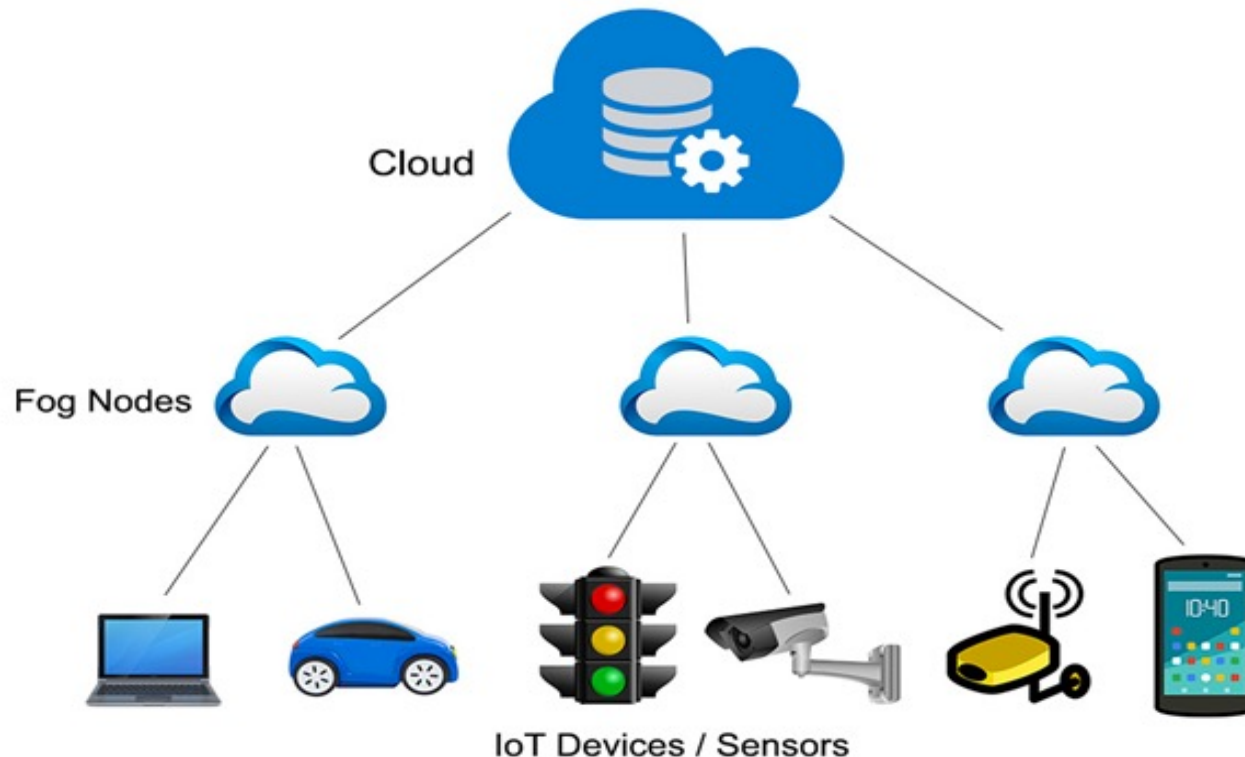
They also facilitate seamless communication between IoT devices through robust APIs and allow interaction between connected devices and smartphones thereby paving the way for the growth of connected technologies.



To new models: Fog Computing

Fog computing or **fog networking**, also known as **fogging**, is an architecture that uses [intermediate sites](#) to carry out a substantial amount of computation ([fog computing](#)), storage, and communication locally and routed over the [Internet](#)

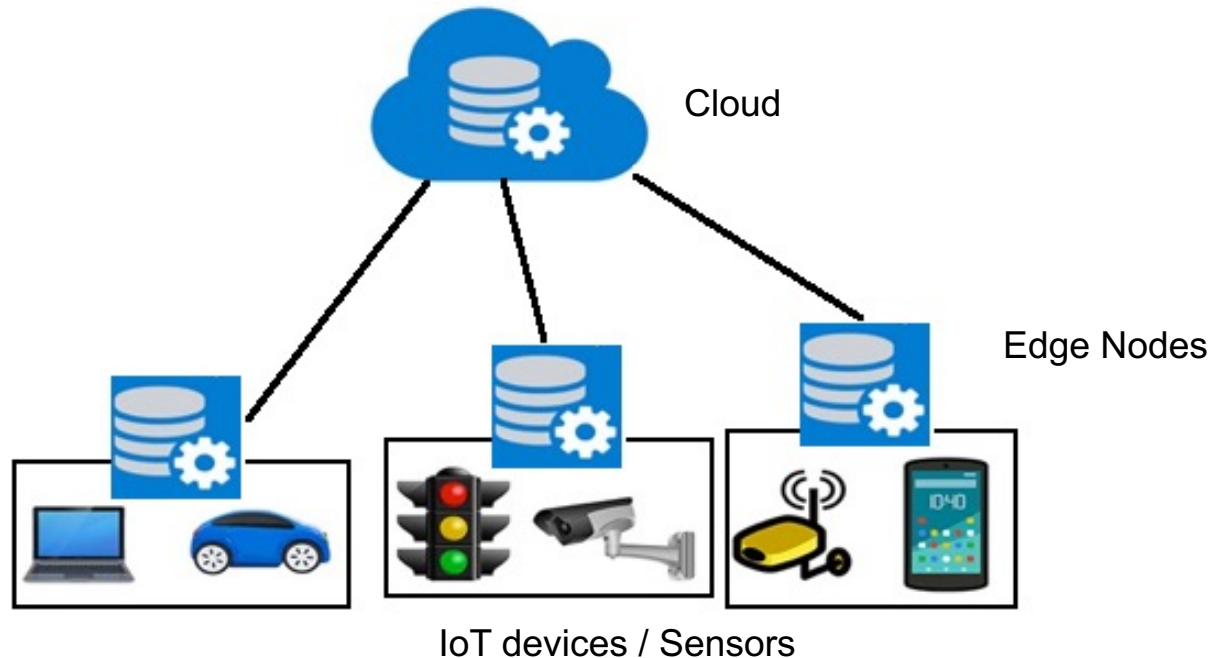
Fogging supports the IoT, in which most of the devices are resource-constrained and have limits in performing heavy computations. A fog site can provide plenty of resources for IoT devices by performing these computations instead



To new models: Edge Computing

Edge computing is a [distributed computing](#) paradigm that brings computation and [data storage](#) closer to the sources of data (our IoT devices).

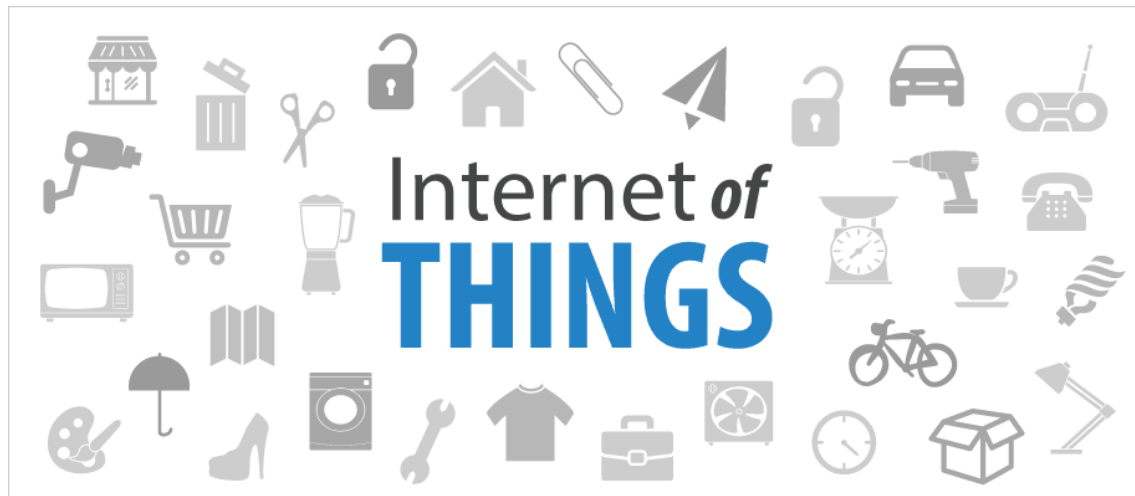
This implies flattening the network to achieve better services and quality of experience by shifting many data collection and processing activities within IoT gateways that, being located near data sources, allow for highly scalable low-latency services.



IoT Definitions

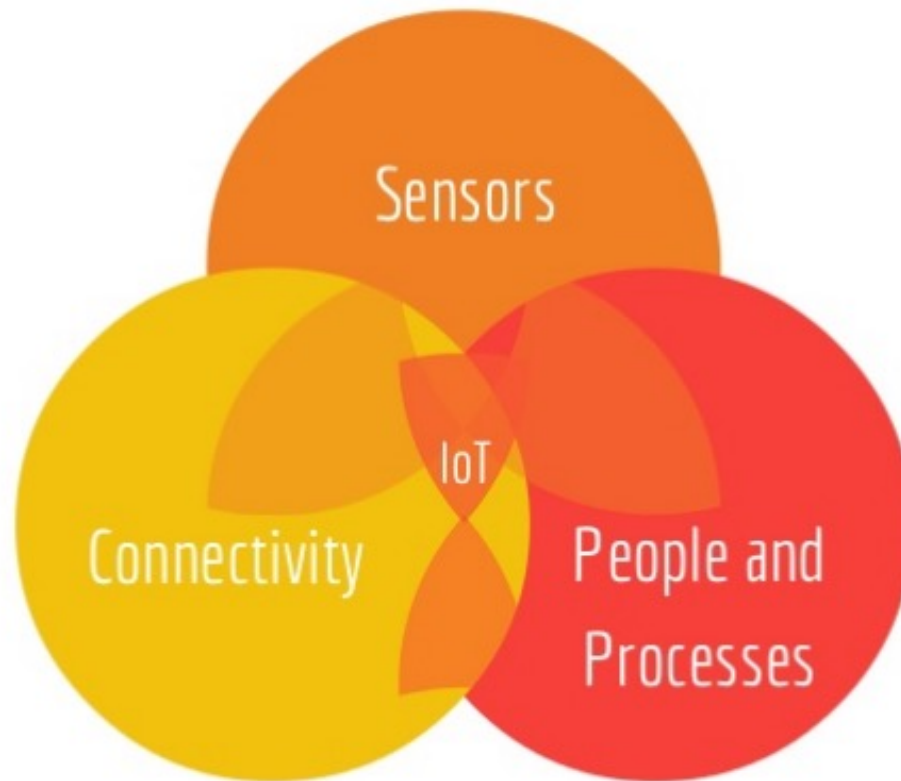
- The **Internet of Things** is a system of physical objects that can be discovered, or interacted with by electronic devices that communicate over various networking interfaces and eventually can be connected to the wider internet.
- A **Thing** may be a sensor or actuator. It may be able to execute computations and or communicate over wired or wireless interfaces.
- A Thing may be **tagged** (passive) or **connected**.

Definitions from "Building the Web of Things" Guinard and Trifa

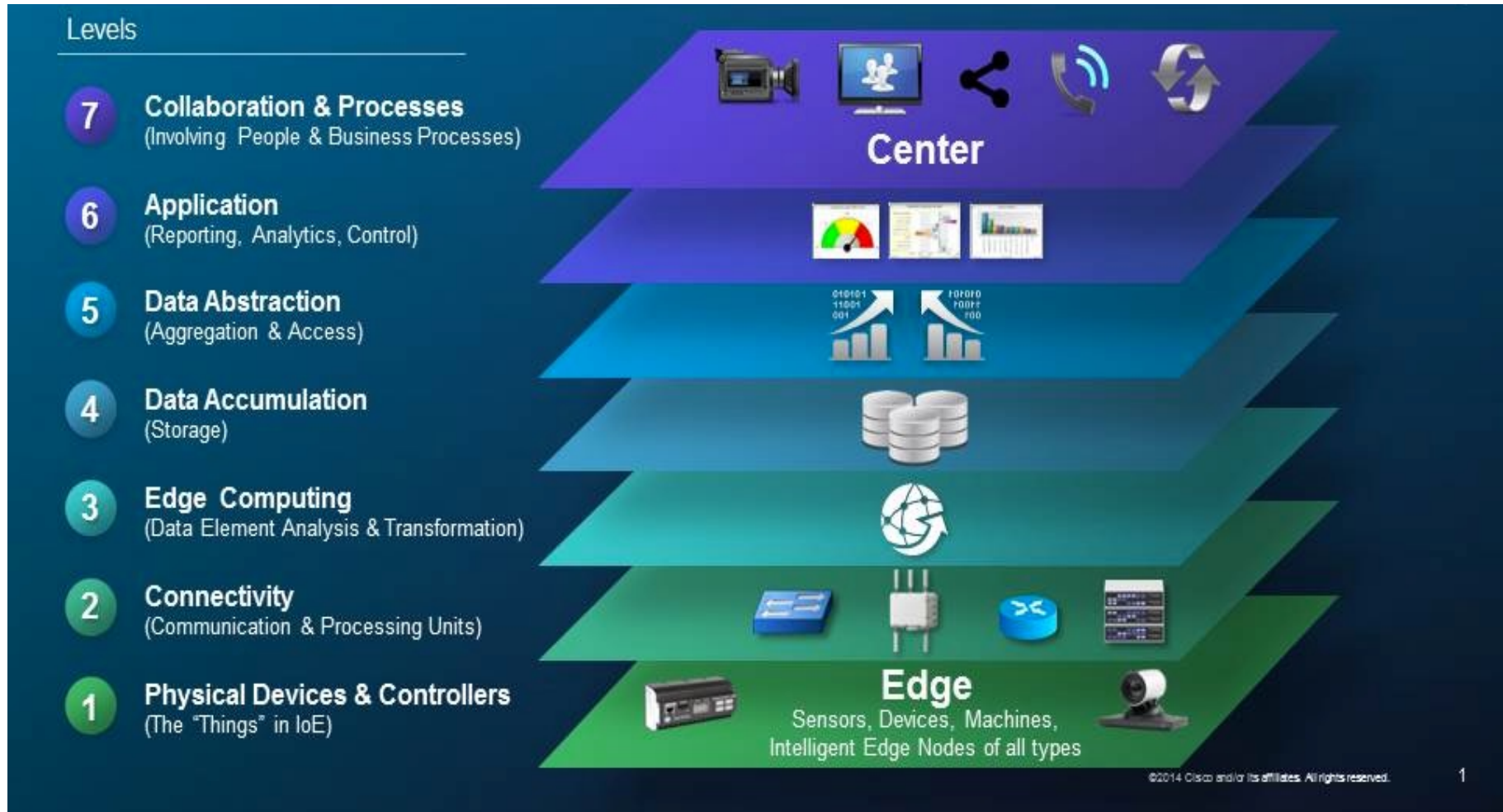


Another view

- The Internet of Things is a combination of:



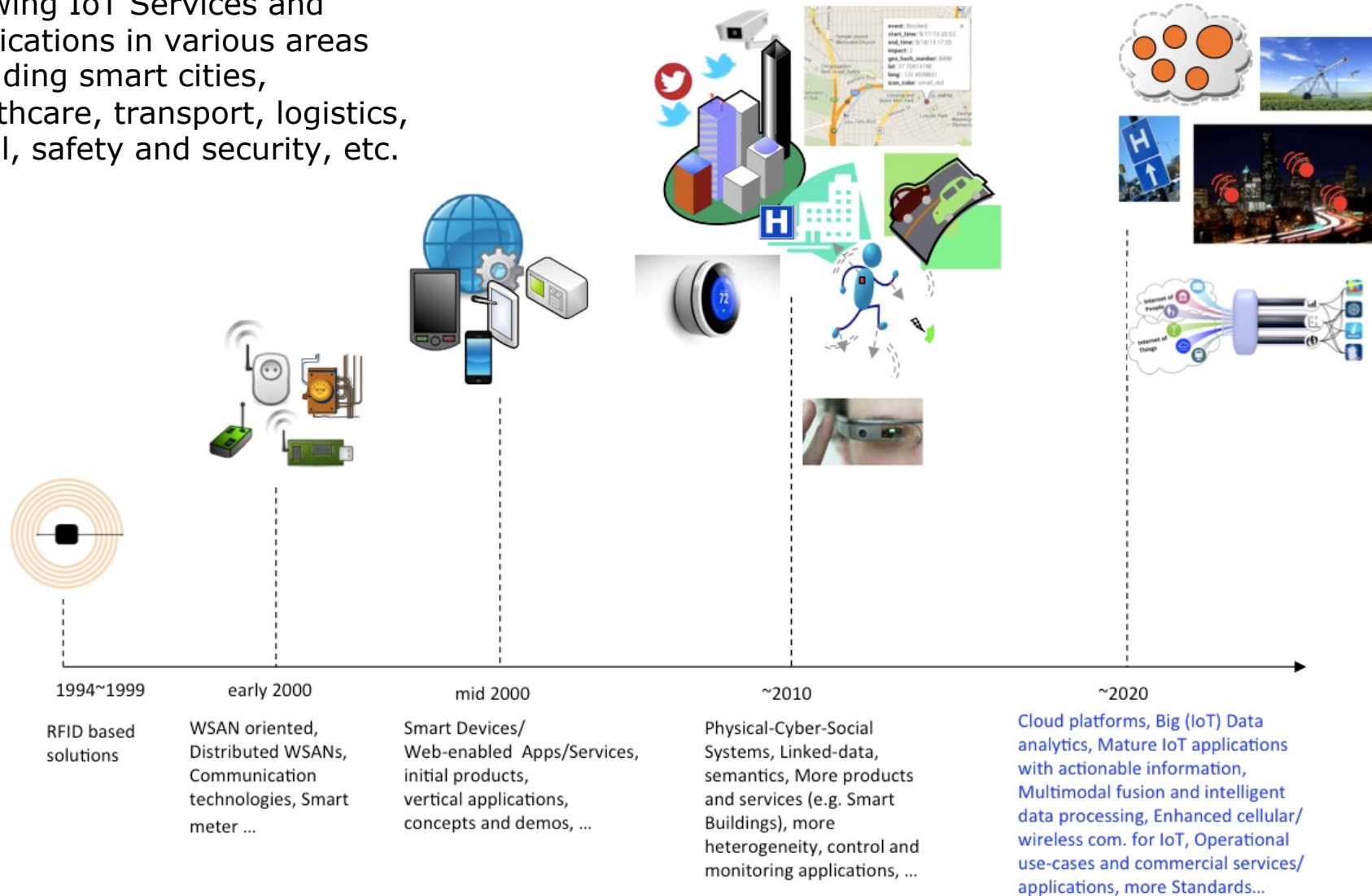
The IoT Reference Model



Source: IoTWF Steering Committee

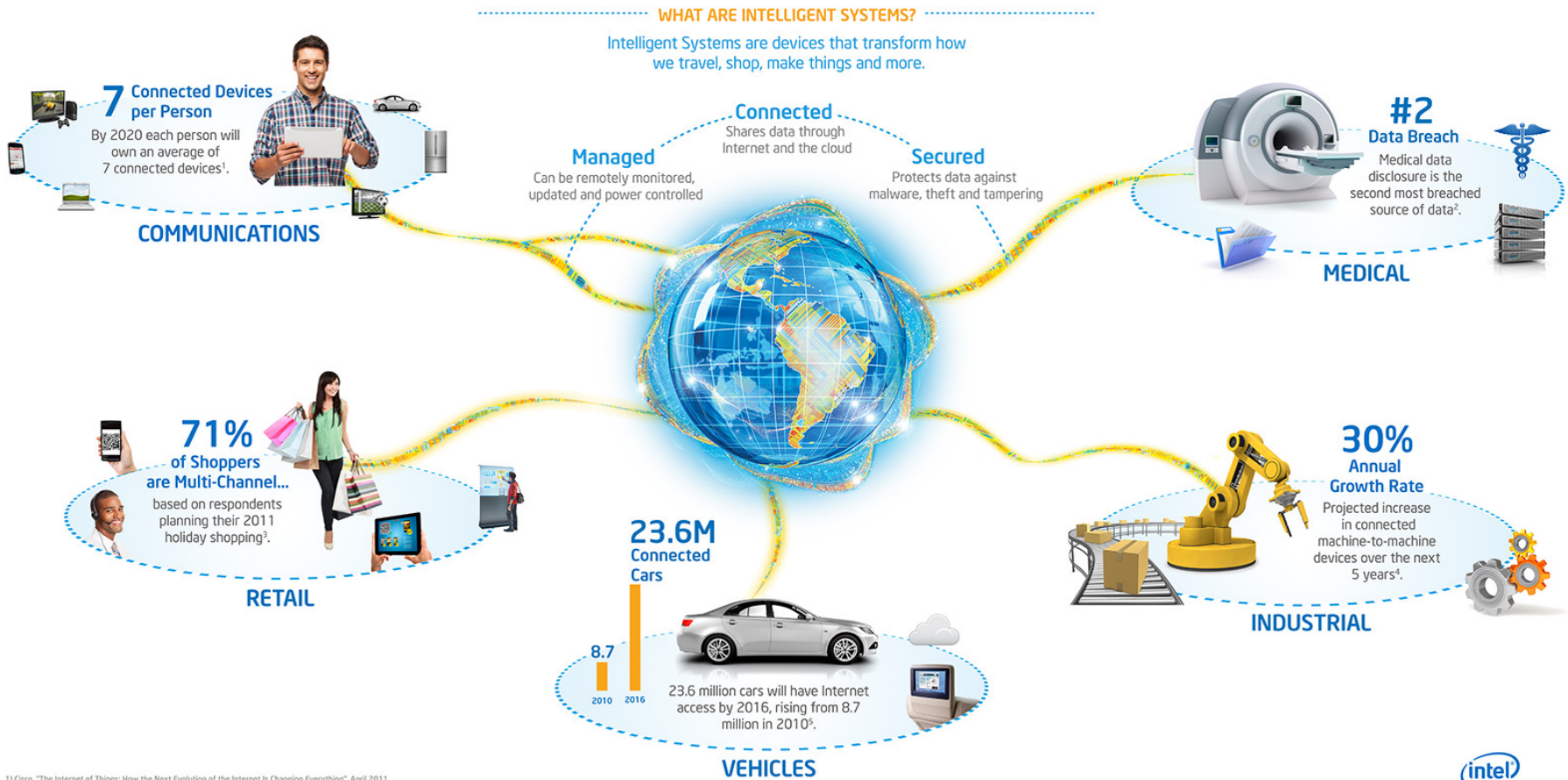
Internet of Things Evolution

Growing IoT Services and Applications in various areas including smart cities, healthcare, transport, logistics, retail, safety and security, etc.



New Opportunities

Intelligent Systems for a More Connected World



1) Cisco, "The Internet of Things: How the Next Evolution of the Internet Is Changing Everything", April 2011

2) Bloor Research, "Security challenges in the US healthcare sector" White Paper, December 2010, <http://www.mcafee.com/us/resources/white-papers/wp-bloor-healthcare-security.pdf>

3) Deloitte U.S., 2011 Annual Holiday Survey, http://www.deloitte.com/assets/Doc-UnitedStates/Local/20Assets/Documents/Consumer%20Business/us_retail_AnnualHolidaySurvey_2011_pr_102611.pdf

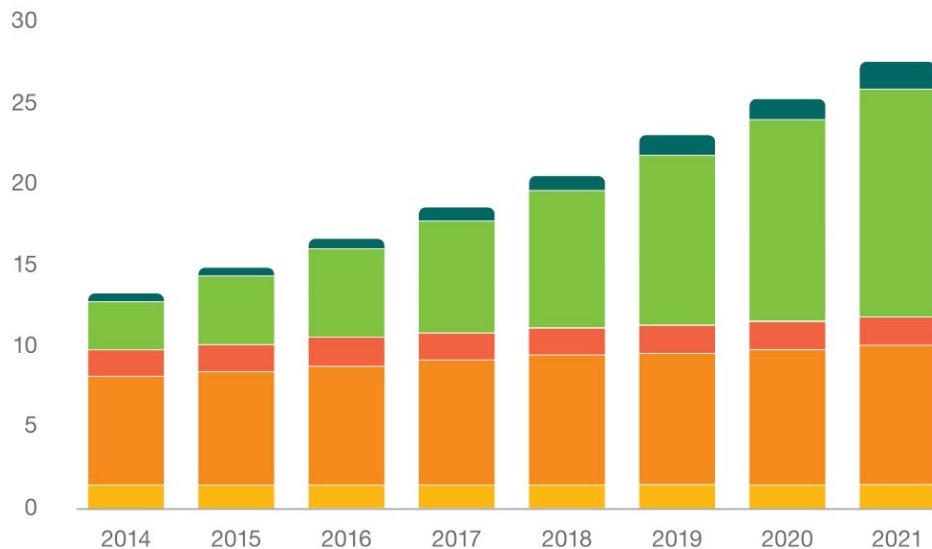
4) McKinsey Global Institute analysis, "Big data: The next frontier for innovation, competition, and productivity", June 2011

5) Wall Street Journal, <http://online.wsj.com/article/SB10001424052702304066504576349763614933844.html>, estimate from research firm, Frost & Sullivan

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IoT dominates the connected devices market

Connected devices (billions)

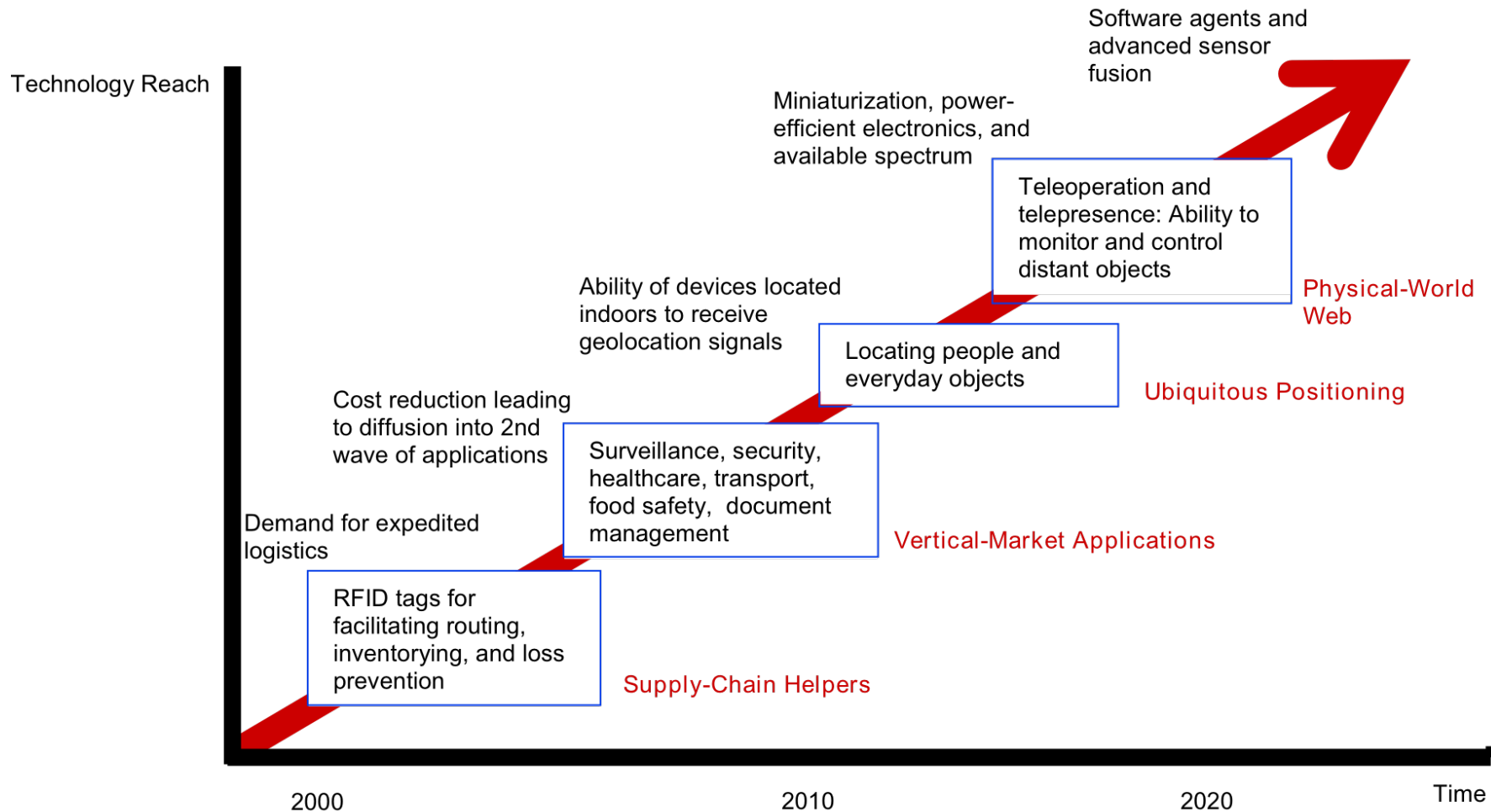


	15 billion	28 billion	CAGR 2015–2021
Cellular IoT	0.4	1.5	27%
Non-cellular IoT	4.2	14.2	22%
PC/laptop/tablet	1.7	1.8	1%
Mobile phones	7.1	8.6	3%
Fixed phones	1.3	1.4	0%

source: Ericsson Mobility Report

Technology trend

TECHNOLOGY ROADMAP: THE INTERNET OF THINGS



Source: SRI Consulting Business Intelligence